5.3.2 Permutation matrices

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Week 05 due Feb 12, 2025 20:19 IST

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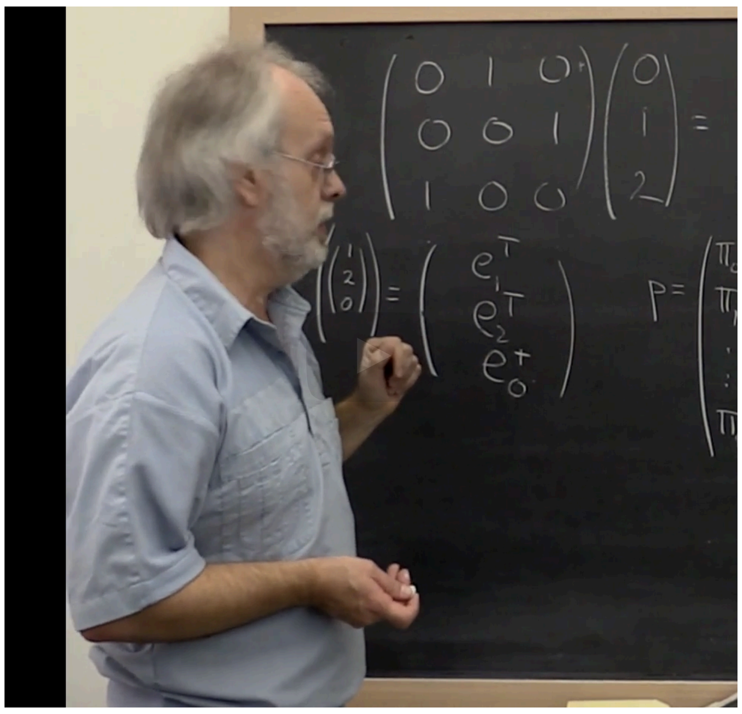

the standard basis vector e1 transposed and then e2 transposed and then e0

transposed. Now, we know that if we multiply this

times a vector, it picks out the entry index with 1, the entry indexed with 2,

the entry indexed with 0. So standard basis vectors can be

used to pick out entries in a vector when you take the dot product. So the

other thing you notice is that if we had only had these indices,

4:14 / 5:03 2.0x

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Reading assignment

1/1 point (graded)

Read Unit 5.3.2 of the notes. [LINK]

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Cost of permutation matrix application?
Suppose there is a matrix A with dimensions n by n and we wish to compute P... 3

? Small typo in solution to HW 5.3.2.4. 2

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vehicle there is going to be something slightly simpler. So when

we get to this point what we want is we want a permutation matrix, and we're

going to call it P tilde, that given an integer,

relative to this vector swaps the row that is relative to this row. So counting

this row as being row 0, this integer tells us which row amongst these rows to

swap. Okay? So that matrix actually has a very special form. That matrix P tilde of

pi, now let's just say

4:28 / 4:28

2.0x

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Homework 5.3.2.1.

1/1 point (graded)

Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix}.$$

Evaluate $P(p)\, x$.

- [ ] $$P(p)\, x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix}$$

- [x] $$P(p)\, x = \begin{pmatrix} \chi_{\pi_0} \\ \vdots \\ \chi_{\pi_{n-1}} \end{pmatrix}$$

Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.3.2.2.

1/1 point (graded)
Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{n-1}^T \end{pmatrix}.$$

Evaluate $P(p)A$.

- [x] $$P(p)A = \begin{pmatrix} \tilde{a}_{\pi_0}^T \\ \vdots \\ \tilde{a}_{\pi_{n-1}}^T \end{pmatrix}$$
- [ ] $$P(p)A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{n-1}^T \end{pmatrix}$$

✔

Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.3.2.3.

1/1 point (graded)
Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = \begin{pmatrix} a_0 & \cdots & a_{n-1} \end{pmatrix}.$$

Evaluate $AP(p)^T$.

- [x] $AP(p)^T = \begin{pmatrix} a_{\pi_0} & \cdots & a_{\pi_{n-1}} \end{pmatrix}$
- [ ] $AP(p)^T = \begin{pmatrix} a_0 & \cdots & a_{n-1} \end{pmatrix}$

✔

Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.3.2.4.

1/1 point (graded)
Evaluate P(p) P(p)^T .

- [x] $P(p)\,P(p)^T = I$
- [ ] $P(p)\,P(p)^T = P(p)$
- [ ] $P(p)\,P(p)^T = P(p)^T$

Solution:

For a complete solution, see the notes .

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